Student Performance Predictor

How to Run Guide

Requirements

Make sure you have Python installed on your computer. You can download it from:

https://www.python.org/downloads/

Setup Steps

Step 1 — Download the project files

Download all these files into one folder on your computer:

- generate_dataset.py
- train.py
- predict.py

Step 2 — Open terminal in VS Code

Open VS Code and use the terminal inside it. Make sure you are inside the project folder. Type this command:

```
cd C:\Users\YourName\performance-predictor
```

Step 3 — Install required libraries

Run this command once and wait for it to finish:

```
pip install pandas scikit-learn
```

Step 4 — Generate the dataset

Run this command:

```
python generate_dataset.py
```

This will create a file called student_data.csv in your folder.

Step 5 — Train the model

Run this command:

```
python train.py
```

This will train the machine learning model and save it as model.pkl. You will also see the accuracy of the model printed on screen.

Step 6 — Run the predictor

Run this command:

```
python predict.py
```

The program will ask you some questions. Type your answers and press Enter. At the end it will show you the predicted score.

What Each File Does

File	What it does
generate_dataset.py	Creates fake student data with 200 records and saves it as student_data.csv
train.py	Reads the CSV file, trains a Linear Regression model and saves it as model.pkl
predict.py	Loads the saved model and asks the user to enter their details then shows the predicted score
student_data.csv	The dataset file created after running generate_dataset.py
model.pkl	The trained model file created after running train.py

Always Run Files in This Order

1. generate_dataset.py
2. train.py
3. predict.py

Troubleshooting

Error	Fix
No such file: student_data.csv	Run generate_dataset.py first before train.py
No such file: model.pkl	Run train.py first before predict.py
Module not found	Run: pip install pandas scikit-learn
Wrong folder error	Make sure terminal is inside the performance-predictor folder